

CO3 ASSESSMENT TOOL2

Case Study Based Assessment

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Source Code (C Program - Merge Sort for Social Media Feed Sorting)

```
#include <stdio.h>

#include <string.h>


#define MAX 100


// Structure to represent a social media post
struct Post
{
    int id;
    long timestamp;
    int relevance;
};


// Merge function
void merge(struct Post posts[], int left, int mid, int right)
{
    int n1 = mid - left + 1;
    int n2 = right - mid;

    struct Post L[MAX], R[MAX];

    int i, j, k;
```

```
for(i = 0; i < n1; i++)  
    L[i] = posts[left + i];
```

```
for(j = 0; j < n2; j++)  
    R[j] = posts[mid + 1 + j];
```

```
i = 0;  
j = 0;  
k = left;
```

```
// Sort by timestamp (latest first)  
while(i < n1 && j < n2)  
{  
    if(L[i].timestamp >= R[j].timestamp)  
    {  
        posts[k] = L[i];  
        i++;  
    }  
    else  
    {  
        posts[k] = R[j];  
        j++;  
    }  
    k++;  
}
```

```
while(i < n1)  
{  
    posts[k] = L[i];  
    i++;  
    k++;  
}
```

```

    }

    while(j < n2)
    {
        posts[k] = R[j];
        j++;
        k++;
    }
}

// Merge Sort
void mergeSort(struct Post posts[], int left, int right)
{
    if(left < right)
    {
        int mid = left + (right - left) / 2;

        mergeSort(posts, left, mid);
        mergeSort(posts, mid + 1, right);

        merge(posts, left, mid, right);
    }
}

// Display Posts
void display(struct Post posts[], int n)
{
    printf("\nID\tTimestamp\tRelevance\n");

    for(int i=0;i<n;i++)
    {
        printf("%d\t%d\t\t%d\n",

```

```

        posts[i].id,
        posts[i].timestamp,
        posts[i].relevance);
    }
}

int main()
{
    struct Post posts[] =
    {
        {101, 1691002300, 85},
        {102, 1691005200, 60},
        {103, 1690999900, 92},
        {104, 1691004300, 75},
        {105, 1691006200, 88}
    };

    int n = sizeof(posts)/sizeof(posts[0]);

    printf("Before Sorting:\n");
    display(posts, n);

    mergeSort(posts, 0, n-1);

    printf("\nAfter Sorting (Latest Timestamp First):\n");
    display(posts, n);

    return 0;
}

```